

Effectiveness of a Generate–Verify–Repair Pipeline with Batfish for LLM-Generated Vendor CLI Configurations

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Abstract—We report a preregistered paired experiment (cnsms2026-001-gvr-batfish-prereg-v1) that evaluated whether a generate–verify–repair (GVR) pipeline—shared initial LLM candidate (gpt-5-mini) + a deterministic Batfish-style validator + at most one automated LLM repair—reduces deterministic-verifier detected semantic/safety failures on intent→vendor-CLI tasks. Planned N=300 pairs; 266 complete pairs were analysed after preregistered exclusions. Baseline pass-rate = 260/266 (97.744%); guarded (final GVR) pass-rate = 265/266 (99.624%). Paired difference = 0.01879699248; exact McNemar two-sided $p = 0.0625$. Paired bootstrap (B=10,000, seed=1729) 95% percentile CI = [0.0037593984962406013, 0.03759398496240601]. Repair was invoked for 6 tasks; median repair iterations per task = 0. Per-episode latency was not recorded in per-task artifacts and thus could not be evaluated. Under shared_initial_candidate semantics the observed absolute improvement is small and did not meet the preregistered $\alpha=0.05$ decision rule. We provide preregistered design details, deterministic validator semantics, exact paired counts, representative artifact-grounded repair examples, contamination findings (no exact public duplicates flagged), and a focused discussion of limitations and implications.

I. INTRODUCTION

Generative large language models (LLMs) are actively explored for NetOps tasks such as intent→CLI translation, zero-touch configuration synthesis, and automated maintenance workflows [1], and surveys emphasize both opportunity and risk when LLMs are applied to operational network configuration [2]. Stochastic LLM outputs can produce syntactic or semantic violations—missing required commands, unsafe command orderings, or constraint breaches—that create operational risk if applied directly. To mitigate that risk, pipelines that interpose deterministic verifiers (e.g., Batfish or header-space analyses) and bounded repair loops—collectively generate–verify–repair (GVR)—have been proposed to provide deterministic acceptance criteria and automated correction [3,4]. Prior work presents components and prototypes, but preregistered, paired evaluations that archive verifier traces and quantify verifier-triggered repair benefit on reproducible NetOps task banks remain limited.

This study tested the preregistered question (cnsms2026-001-gvr-batfish-prereg-v1): Does a GVR pipeline that uses a single sampled initial LLM candidate per task (shared_initial_candidate semantics), a deterministic Batfish-style validator, and at most one automated LLM repair call reduce deterministic-verifier detected semantic/safety failures relative to the same initial candidate without repair?

Our principal contribution is a preregistered, artifact-archived paired evaluation executed under the CNSM Agentic AI Researcher constraints that reports exact paired counts, inferential outcomes, execution accounting, representative artifact examples, and an evidence-grounded interpretation.

II. RELATED WORK

LLM applications in NetOps include intent→CLI translation systems and focused benchmarks studying automated configuration synthesis and correctness [1]. Surveys synthesize common failure modes (including hallucination and constraint violations) and advocate grounding and verification components for automation [2]. Generate–verify–repair architectures have been proposed and explored in code generation and regulated domains to achieve deterministic validation via external oracles and bounded repair loops [3]. Network verification tools such as Batfish and header-space analyses provide deterministic reachability and ordering checks and are natural verifier choices in LLM-augmented NetOps prototypes; prior analyses document Batfish’s design and motivate its integration in automated pipelines [4]. Work examining what LLMs need to synthesize correct router configurations highlights sequencing and vendor-syntax difficulties that verifiers can detect and flag for repair [5].

Our study differs by executing a preregistered paired test that (a) uses shared_initial_candidate semantics exactly as registered, (b) archives raw model and validator traces for auditability, and (c) reports exact paired contingency counts together with the preregistered exact McNemar test and a paired bootstrap CI to aid interpretation in a small-discordant sample.

III. METHODOLOGY

Preregistration and estimand: The study preregistration is cnsms2026-001-gvr-batfish-prereg-v1 (manifest SHA-256: 46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891). Primary estimand: paired_success_rate_difference_guarded_minus_baseline under shared_initial_candidate semantics. Under those semantics the identical single sampled initial model candidate is used to compute both outcomes: baseline single-shot candidate → deterministic validator pass/fail, and guarded outcome → same candidate; if it fails, the guarded arm may invoke at most one automated LLM repair call and the final

validator pass/fail is recorded. This isolates the effect of the allowed validator-triggered repair step for a fixed initial candidate.

Task bank and sampling: The preregistered task bank deterministically produced 300 intent→CLI pairs (150 Cisco-like; 150 Juniper-like) using a seeded synthetic task generator combined with public NetConfEval-style examples; per-task generator ids, seeds, and manifest entries are archived. Inclusion/exclusion, retry, and missingness rules followed the preregistration: provider/API transient failures allowed up to two automatic retries; pairs where both arms failed due to provider/infrastructure errors were excluded from the primary analysis. The execution manifest records `planned_episode_count = 600` and `completed_episode_count = 532`; missingness and reconciliation artifacts are archived.

Model, orchestration, and sampling disclosure: The declared model was `gpt-5-mini` (execution/model_configuration.json field `declared_model_version = "gpt-5-mini"`). Archived configuration fields include `max_output_tokens = 1024`, `maximum_attempts_per_call = 1`, `provider = openai_responses`, and `temperature = null` (unset). Provider accounting (deterministic_reconciliation.json) records `hosted_model_call_event_count = 306`, `completed_hosted_model_call_event_count = 272`, and `failed_hosted_model_call_event_count = 34`. Because temperature was unset and no centralized deterministic sampling seed was archived for provider calls, nondeterministic sampling cannot be excluded for recorded model calls; we note this explicitly in the Disclosure and Limitations.

Deterministic validator semantics and scoring rules: The binary oracle used across episodes was deterministic_netops_validator_v5. For each candidate the validator performed: (1) CLI syntactic parsing/normalization; (2) per-task required-sequence checks (expected_sequence vs parsed commands); (3) constrained-field touch checks (allowed_fields, allowed_touched_fields); and (4) reachability/constraint validation on the synthetic topology model. Each episode archives validation_before and validation_after traces containing parsed_command_count, required_command_count, observed_touched_field_count, violation codes, and normalized_configuration; the validator’s pass/fail output is the primary binary endpoint. The confirmatory scoring rule was ‘full_intent_constraint_validation’ as recorded in analysis artifacts.

Repair loop mechanics (bounded, archived): Per the preregistration the repair stage is bounded to at most one automated LLM repair call per task when the initial candidate fails the validator. The repair prompt includes validator failure codes and task constraints and requests a corrected configuration; repair provider traces and repair_prompt_sha256 are archived when invoked. Stage accounting shows `shared_initial_generation = 300` and `repair = 6` (six repair calls executed). Repair artifacts and validator traces for repaired episodes are included in execution/scoring and representative artifacts described below.

Analysis plan and inferential methods: The prespecified primary test is the exact McNemar two-sided test on paired binary outcomes at $\alpha=0.05$. Interval estimation was preregistered as a paired bootstrap percentile CI ($B=10,000$, $\text{seed}=1729$) for the paired difference to quantify magnitude and uncertainty. Missingness handling followed the preregistered ‘complete_pair_primary’ rule: both-arm provider failures were excluded from the primary test; archived sensitivity analyses treat excluded episodes conservatively as failures. Secondary estimands included median_repair_iterations_per_task and median end-to-end latency; per-episode latency was not archived and thus could not be evaluated. We included the paired bootstrap CI alongside McNemar because exact McNemar p-values can be discrete and conservative with few discordant pairs, whereas a bootstrap CI summarizes empirical paired-difference variability (see Results for the observed small-discordant behavior).

IV. RESULTS

Execution accounting and sample construction. The preregistered run planned 600 model-condition episodes ($300 \text{ tasks} \times 2 \text{ conditions}$). The orchestrated pipeline completed 532 episodes and recorded 68 failed episodes (execution_manifest.json). After applying preregistered exclusion rules that remove pairs affected by both-arm provider/infrastructure failures, the confirmatory analysis used 266 complete paired units (analysis/missingness_summary.csv). Key provider-call accounting from the deterministic reconciliation artifact: `hosted_model_call_event_count = 306` total model call events, of which 272 completed successfully and 34 failed; cache was unused (`cache_hit_count = 0`, `cache_miss_count = 272`). Stage accounting shows that a shared initial generation was produced for every task (`shared_initial_generation = 300`) and the automated repair stage executed in 6 episodes (`repair = 6`). All execution-level manifests and per-episode traces are archived to permit replay and audit.

Primary paired outcomes and exact counts. The confirmatory paired contingency (guarded vs baseline) computed on the 266 complete pairs is shown in analysis/paired_contingency_table.csv and summarized here: $n_{11} = 260$ (both baseline and guarded pass), $n_{10} = 5$ (guarded pass, baseline fail), $n_{01} = 0$ (baseline pass, guarded fail), $n_{00} = 1$ (both fail). These margins yield observed success rates $\text{baseline} = 260/266 = 0.9774436090$ and $\text{guarded} = 265/266 = 0.9962406015$ with an absolute paired difference of 0.01879699248.

Inferential results: exact McNemar and paired bootstrap CI. Per the preregistered primary test, the exact McNemar two-sided p-value for the observed discordant counts ($n_{10}=5$, $n_{01}=0$) is $p = 0.0625$, which does not meet the preregistered $\alpha = 0.05$ decision rule for rejection. To quantify effect magnitude uncertainty we computed the preregistered paired bootstrap percentile 95% CI with $B = 10,000$ resamples and $\text{seed} = 1729$ (analysis/analysis_log.jsonl); the resulting CI is $[0.0037593984962406013, 0.03759398496240601]$, narrowly

excluding zero. Reporting both results preserves the preregistered decision rule while also conveying interval evidence: the discrete nature of McNemar’s exact two-sided test with few discordants can produce conservative p-values even when bootstrap resampling indicates a small positive paired difference with limited sampling variability. Both inferential artifacts (exact p and bootstrap CI with documented seed and resamples) are archived.

Repair invocation behavior and representative repaired episodes. The repair stage executed for 6 of the 300 tasks (stage_counts: repair = 6). Median repair iterations per task across the analysed sample is 0 (majority of tasks required no repair because the shared initial candidate already passed validation). Representative repaired episode: task-000008. The shared initial candidate for task-000008 (execution/responses/task-000008-shared-initial.txt) contained a truncated final token and failed validator checks (validation_before showed parsed_command_count < required_command_count). The automated repair call produced a corrected candidate (execution/responses/task-000008-guarded.txt) whose validator trace (execution/scoring/task-000008-guarded.json) records validation_after.valid = true with no violation codes. Per-episode scoring JSONs record whether a repair was applied, the repair_prompt reference, and the final normalized_configuration accepted by the deterministic validator; all such per-episode repair traces are archived for examination.

Contamination and sensitivity to missingness. The preregistered contamination detector flagged no exact public duplicates (analysis/contamination_summary.csv empty), so no tasks were excluded from the primary analysis on contamination grounds. The preregistered missingness policy excluded 34 both-arm failed episodes from the primary test; a sensitivity analysis that conservatively treats those 34 excluded pairs as failures (i.e., complete N = 300 with both arms set to failure where missing) yields baseline = 260/300 = 0.866667 and guarded = 265/300 = 0.883333, but retains the same discordant structure (n10=5, n01=0) leading to the same exact McNemar p = 0.0625. Thus, under the preregistered conservative missingness treatment the formal decision does not change; archived missingness_summary.csv and deterministic_reconciliation.json provide the complete accounting.

Supplementary quantitative artifacts and reproducibility. Analysis artifacts included in the evidence bundle and referenced above provide the reproducible numeric foundations for these results: analysis/condition_summary.csv (baseline: 260 successes, 6 failures; guarded: 265 successes, 1 failure), analysis/paired_contingency_table.csv (n11, n10, n01, n00 table), analysis/missingness_summary.csv (counts of excluded and failed episodes), and analysis/analysis_log.jsonl (bootstrap settings: B=10,000, seed=1729). deterministic_reconciliation.json documents provider call totals and stage counts that explain model_calls_used = 306 and repair_calls = 6. All files and their hash manifests are archived to enable exact reanalysis or targeted follow-up audits.

V. DISCUSSION

Statistical interpretation and the role of small discordant counts: The paired analysis observed only five discordant pairs favoring the guarded pipeline (n10=5) and zero discordant pairs in the opposite direction (n01=0). Exact McNemar two-sided testing evaluates the symmetry of discordant counts and, for small integer discordant counts, yields discrete p-values; with (b,c)=(5,0) the exact two-sided p = 0.0625 as archived. The preregistered paired bootstrap percentile CI (B=10,000, seed=1729) for the paired difference produces a 95% interval that narrowly excludes zero [0.00376, 0.03759]. Both results are reported because they illuminate different aspects of the same data: the exact McNemar p emphasizes a strict null-hypothesis test under paired discrete outcomes, while the bootstrap CI summarizes empirical uncertainty about the magnitude of the paired difference under resampling of the observed complete pairs (bootstrap settings and seed are archived in analysis_log.jsonl). The observed apparent tension (p>0.05, CI excluding zero) arises from discreteness and small discordant counts rather than from contradictory raw facts; readers should interpret the pair together: a small positive effect was observed, but it did not meet the preregistered exact-test decision threshold.

Effect size, power, and practical detectability in context: The preregistered power calculation targeted detecting a 0.10 absolute improvement at 80% power with N=300 pairs. The observed effect (≈ 0.0188) is far smaller than that target; with complete_pairs = 266 (reduced from planned 300 due to 34 both-arm provider failures) the realized power to detect such a small effect is negligible. In practical terms this experiment demonstrates that when baseline first-pass success is already high ($\approx 97.7\%$), bounded automated repair that can be invoked at most once per failed candidate will rarely be exercised (6 repairs invoked across 300 planned tasks) and thus can only produce small absolute improvements unless the baseline error rate is larger or the repair stage is more permissive (e.g., allow multiple repair iterations or stronger repair strategies).

Artifact-grounded diagnostics of pipeline failure modes: The archived per-episode validator traces permit direct inspection of why candidates failed and how repairs corrected them. Representative diagnostics include: - Truncation/malformed output: task-000008’s shared initial candidate shows a truncated terminal line ("interface uplink2\n") that caused parsed_command_count mismatches; the automated repair produced the missing explicit admin down line and then passed the validator. Repair artifacts record both the validator violation codes and the repair prompt used, enabling replay of the exact repair stimulus. - Required-sequence violations: several failure traces indicate missing or misordered commands relative to the task’s required_sequence; these are precisely the semantic checks the deterministic validator enforces and which repair prompts referenced when invoked. - Constrained-field touch checks: validator traces include observed_touched_field_count and allowed_touched_fields enforcement; failures of this type reflect field-scope violations

rather than pure syntax and are detectable deterministically by the validator. These concrete, archived diagnostics show that the validator’s rule battery produced meaningful, actionable failure signals that the repair prompt could address in some cases, but also that many initial candidates already satisfied the rule battery and thus were unaffected by the repair stage.

Operational and design implications grounded in execution artifacts: The execution accounting (deterministic_reconciliation.json and provider call logs) shows: `hosted_model_call_event_count` = 306, `completed_hosted_model_call_event_count` = 272, `failed_hosted_model_call_event_count` = 34, `shared_initial_generation` = 300, and `repair` = 6. This instrumentation supports several concrete operational inferences: 1) Low marginal utility of bounded repairs on high-accuracy corpora: with few initial failures, a guarded repair stage invoked at most once yields a small absolute improvement; this suggests teams evaluating GVR in practice should calibrate repair budget (number of allowed repair iterations, repair prompt sophistication) to baseline error rates. 2) Auditability and triage: the archived validator traces and scoring JSONs provide a compact, machine-readable audit trail suitable for operator triage and for post-deployment incident analysis; these artifacts concretely demonstrate how deterministic validation can localize a failure cause (syntax vs sequence vs field scope). 3) Latency and throughput unknowns: per-episode timing was not archived and therefore the preregistered latency estimand could not be evaluated; absent timing data, operational latency tradeoffs between invoking a repair and manual review remain unmeasured in this run.

Threats to validity revisited with artifact evidence: Internal validity: the study respected `shared_initial_candidate` semantics preregistered intent and used identical initial candidates for both arms; as a result, the measured estimand isolates the effect of the validator-triggered repair only. Construct validity: `deterministic_netops_validator_v5` enforces a limited, archived rule battery (syntactic parsing, required_sequence checks, constrained_field checks, reachability on the synthetic topology). Semantic failures outside that battery (for example, subtle policy violations not encoded in the validator) would go undetected and are thus absent from the measured endpoint. External validity: the task corpus combined public NetConfEval-style examples and a seeded synthetic generator limited to two vendor dialect clusters; the manuscript therefore does not claim generality to other vendors, larger topologies, or production heterogeneity. Conclusion validity: provider failures reduced the analyzed N (266 complete pairs) and the small number of discordant pairs limits inference for small effects.

Concrete recommendations for future experiments (artifact-grounded): The archived manifests suggest the following preregistered changes would increase diagnostic value and power in follow-on studies: (a) design arms that manipulate first-pass generation (e.g., constrained prompting, multiple independent draws) rather than relying

solely on `shared_initial_candidate` semantics if the goal is to measure effects of prompting or sampling; (b) allow a bounded sequence of repair iterations (≥ 1) and instrument repair-step efficacy; (c) record per-episode latency (request→generation→validation→repair→final validation) to enable formal latency/operational tradeoff analysis; (d) expand and publish the validator rule battery and add adversarial synthetic tasks specifically crafted to probe known verifier false negatives—per-episode validator traces in this run show that such audits are feasible; and (e) if the target operational environment has high baseline accuracy, power calculations should target smaller effect sizes or alternative estimands (e.g., time-to-triage reduction, human-in-the-loop workload) that better reflect marginal benefits of repair.

Reproducibility and archival diagnostics: All raw responses, provider call traces, validator traces, scoring JSONs, analysis artifacts (paired contingency table, condition summary, missingness summary, contamination summary), `deterministic_reconciliation.json`, `model_configuration.json`, and `analysis_log.jsonl` (bootstrap parameters: $B=10,000$, $seed=1729$) are archived and hashed in the evidence bundle. These artifacts permit exact reanalysis of the paired contingency, replay of repair prompts with the same validator failure codes, and targeted inspection of individual failure modes. The provider accounting and stage counts enable independent auditors to confirm that the repair stage executed only six times and that 34 provider calls failed and were excluded per preregistered rules.

Synthesis: The experiment provides an artifact-rich, preregistered measurement of the limited marginal benefit of a tightly bounded GVR pipeline under `shared_initial_candidate` semantics when baseline first-pass accuracy is high. The archived traces make clear which failure modes the validator detected, which were amenable to one automated repair call, and where further engineering (expanded validator coverage, richer repair strategies, or different experimental arms) would be required to observe larger absolute benefits.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics: both arms used the same single sampled initial candidate per preregistration; the estimand measures validator-triggered repair benefit for that fixed initial sample and does not evaluate independent first-pass generations or constrained first-pass prompting strategies.
- Missingness and sample reduction: planned $N=300$ pairs; after infrastructure/provider exclusions 266 pairs were complete and 34 both-arm episodes were excluded per preregistered rules, reducing effective sample size and precision.
- Small discordant counts: primary inference used exact McNemar with few discordants ($n_{10}=5$, $n_{01}=0$); conclusions are sensitive to inferential choice and limited power.
- Validator coverage: `deterministic_netops_validator_v5` enforces a finite set of syntactic/semantic/reachability

rules; semantic violations outside its coverage are possible and unmeasured.

- Runtime nondeterminism and sampling: archived execution/model_configuration.json records temperature = null and no deterministic seed; nondeterministic sampling cannot be excluded for recorded model calls.
- H2 latency not evaluated: per-episode end-to-end latency was not present in archived per-task artifacts and therefore the preregistered latency bound (median ≤ 60 s) could not be assessed.
- External validity: task corpus derives from public NetConfEval-style examples and a seeded synthetic generator limited to two vendor-dialect clusters; results do not imply universal benefit across other vendors, larger topologies, or production deployments.

VII. CONCLUSION

We executed a preregistered paired experiment that evaluated a GVR pipeline (gpt-5-mini + deterministic_netops_validator_v5 + ≤ 1 automated repair) on intent $\rightarrow$ CLI tasks under shared_initial_candidate semantics. Across 266 analysed pairs the guarded pipeline improved validator pass rate by 0.01879699248 but did not meet the preregistered exact-McNemar $\alpha=0.05$ decision rule ($p=0.0625$); a paired bootstrap CI ($B=10,000$, $\text{seed}=1729$) narrowly excludes zero. Repair calls were rare (6/300) and median repair iterations per task = 0. Per-episode latency was not archived and could not be assessed. All execution and analysis artifacts (raw responses, validator traces, provider call accounting, and reconciliation manifests) are archived in the evidence bundle to support reanalysis and follow-on work.

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DISCLOSURE STATEMENT

Disclosure (concise): Declared model: gpt-5-mini (execution/model_configuration.json archived; declared_model_version = "gpt-5-mini"). Orchestration

adapter: hosted_netops_gvr_v1. Deterministic validator: deterministic_netops_validator_v5. Preregistration: cnsms2026-001-gvr-batfish-prereg-v1 (manifest SHA-256: 46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891). Initial master prompt immutable reference: provenance/master_prompt.txt (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba). Human role and autonomy boundary: before the autonomy lock a human Research Director selected the topic family and approved pre-lock infrastructure development; after the autonomy lock the registered pipeline autonomously performed literature selection, preregistration, experimental execution, analysis, manuscript drafting, AI peer review simulation, and revision. Nondeterminism disclosure: archived execution/model_configuration.json records temperature = null (unset) and no deterministic sampling seed is archived; therefore nondeterministic sampling of model outputs cannot be excluded for the recorded provider calls. Execution and analysis artifacts, logs, and hash manifests are archived in the evidence bundle referenced in the preregistration.

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